

Xiaoxia Gao

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Personal Statement

- I am a Master's student at Xiamen University focusing on wearable systems and human-computer interaction. My goal is to develop embodied technologies that turn real-world sensory experience into adaptive, deployable systems for everyday human benefit.

Education

09/2024 - Present

Xiamen University, Fujian, China

- Master of Computer Science and Technology, School of Informatics**
- Research Interests: wearable systems, human motion capture and human-computer interaction

09/2020 - 07/2024

Xidian University, Xian, China

- Bachelor of Computer Science and Technology**
- GPA: 3.80/4.00 Integrated Rank: 15/133

Publication

- Xiaoxia Gao**, Ruonan Zheng, Qinze Zhou, Yu Yan, Zixuan Zeng, Anjun Chen, Yipeng Qin, Shihui Guo, Jiawei Fang, Foundation Garment Poser: An Open Foundation Model for Garment-based Inertial Motion Capture. Under review at Siggraph Asia 2026.
- Jiawei Fang*, Ruonan Zheng*, **Xiaoxia Gao***, Shifan Jiang, Anjun Chen, Qi Ye, Shihui Guo. Garment Inertial Denoiser: Endowing Accurate Motion Capture via Loose IMU Denoiser. Under review at ACM Multimedia 2026.
- Xiaoxia Gao***, Jiawei Fang*, Yipeng Qin, Shihui Guo, Yiyue Luo, Learning Different Body Shape, Wearing Position and Joint Angle from Strain Curve. Plan to submit to Nature Machine Intelligence.
- Jiawei Fang*, Ruonan Zheng*, Yuan Yao, **Xiaoxia Gao**, Chengxu Zuo, Shihui Guo, Yiyue Luo, FIP: Endowing Robust Motion Capture on Daily Garment by Fusing Flex and Inertial Sensors, arxiv. CHI 2025.
- Jiawei Fang, Haishan Song, Chengxu Zuo, **Xiaoxia Gao**, Xiaowei Chen, Shihui Guo, Yipeng Qin, SuDA: Support-based Domain Adaptation for Sim2Real Motion Capture with Flexible Sensors. ICML 2024.

Research Experience

12/2025 - 05/2026

Foundation Garment Poser: An Open Foundation Model for Garment-based Inertial Motion Capture

- Contribution:** Propose FGP, an open foundation model for garment-based inertial motion capture which enables a single shared model to handle varying IMU counts, sensor layouts, and garment configurations without configuration-specific retraining.
- Approach:** Designed FGP, a two-stage foundation model factorizing garment MoCap into IMU Canonicalization and Motion Reconstruction. A Masked IMU Encoder pretrained on large-scale unlabeled garment data provides configuration-invariant representations; a latent-conditioned IMU Canonicalizer then leverages limited paired supervision to recover canonical tight-wear IMU signals, enabling the downstream pose estimator to reuse pose priors learned from large-scale synthetic data.
- Achievement:** Collect the largest garment-based inertial Mocap dataset to date, containing about 101.1 hours unlabeled garment data and 15 hours paired garment-motion data across five garment designs, and 26 participants. Outperform the accuracy of state-of-the-art (42%↑), zero-shot cross-garment generalization, long-term robustness, and favorable scaling behavior, while supporting practical applications such as in-the-wild motion capture and humanoid teleoperation.

03/2025 - 06/2026

Garment Inertial Denoiser: Endowing Accurate Motion Capture via Loose IMU Denoiser

- Contribution:** Proposed Garment Inertial Denoiser (GID), a lightweight plug-and-play Transformer module that enables accurate full-body motion capture from loose garment-mounted IMUs.
- Approach:** Designed a factorized loose-to-tight IMU denoising pipeline with location-specific Transformer experts and adaptive cross-wear fusion.
- Achievement:** Collected GarMoCap and achieved strong full-body MoCap performance using only 11 minutes of single-user training data, improving ASIP angular error from 26.09° to 15.68° while generalizing to unseen garments and users.

09/2024 - Present

Wear-It-Free: Dynamic Calibration Joint Tracking on Flexible Sensors with Motion-Invariant Strain Surface

- Contribution:** Introduced the first imperceptible calibration framework for flexible-sensor-based joint tracking, enabling robust tracking across body shapes, wearing positions, motions, and device configurations.
- Approach:** Formalized this factorization as a Motion-Invariant Strain Surface (MISS) over hinge joints, and built WIFE, a lightweight framework that performs imperceptible real-time calibration via strain matching against MISS, eliminating the need for explicit user calibration while continuously compensating for sensor displacement during extended wear.

- **Achievement:** Built and evaluated WIFEVAL, a large-scale benchmark with 130 participants, 172 wearing positions, 10 motions, and 11 devices; outperformed SOTA methods by 57% error reduction on WIFEVAL with only 1,654 training frames / 33s, 0.03s calibration, and real-time inference.

Awards & Honors

- National Encouragement Scholarship (2021, 2022)
- University-Level Outstanding Student (2021)
- Meritorious Winner in the Mathematical Contest in Modeling (MCM/ICM, USA) (2022)
- Third Prize in College English Competition (C Level) (2022)

Academic Skills

- **Software and Tools:** Origin, Blender, Unity, Marvelous Designer, XINGYING optical motion capture system, 8GBR camera-array motion capture system, and Noitom inertial motion capture system.
- **Language:** Mandarin (Native), English (Proficiency)
- **Technical:** Python, C, C++, LaTeX